

EPICS AND USER STORIES

EPIC 1: Data Foundation & Architecture

Goal: Build a clean, analysis-ready data warehouse to support predictive modeling and dashboards.

User Story 1.1 — Create Centralized Data Warehouse

As a Data Engineer

I want to build a Snowflake data warehouse following the star schema

So that the analytics and model pipeline have a single source of truth.

Acceptance Criteria

- Snowflake environment created and tested
 - Five core tables implemented: patients, providers, clinics, appointments, dates
 - Data dictionary drafted and reviewed
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User Story 1.2 — Implement Star Schema

As a Data Engineer

I want a clear relational structure between fact and dimension tables

So that queries are fast, consistent, and ready for BI tools.

Acceptance Criteria

- Dimension tables populated
 - Fact table (appointments) links to all dimension keys
 - Schema validated by BA + DS
 - All joins tested for correctness
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User Story 1.3 — Perform Data Quality Assessment

As a Business Analyst

I want clarity on data completeness, accuracy, and gaps

So that the team knows constraints that impact the predictive model.

Acceptance Criteria

- Missing data statistics reported

- Anomalies or inconsistencies flagged
 - DE provides remediation plan
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EPIC 2: Predictive No-Show Model Development

Goal: Build a baseline no-show prediction model with engineered features and a functioning API endpoint.

User Story 2.1 — Engineer Model Features

As a Data Scientist

I want to create historical, demographic, and scheduling-based features

So that the model can learn key no-show patterns.

Acceptance Criteria

- Features generated: patient history, temporal flags, lead time, insurance type
 - Feature dataset validated against business rules
 - Ready for training by Day 3
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User Story 2.2 — Train Baseline No-Show Model

As a Data Scientist

I want a baseline predictive model using scikit-learn

So that we can measure initial performance and iterate.

Acceptance Criteria

- Baseline model trained
 - Performance metrics documented (AUC, precision, recall)
 - Model card created
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User Story 2.3 — Build FastAPI Endpoint

As a Developer

I want a deployed API that returns predictions in <300ms

So that operational systems can integrate seamlessly.

Acceptance Criteria

- API accepts structured input
- Returns prediction + risk score

- Average response time <300ms
 - Fully documented endpoint
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EPIC 3: Patient Flow & Operational Analytics

Goal: Build interactive dashboards enabling clinics to act on data-driven insights.

User Story 3.1 — Build KPI Cards (No-Show, Wait Time)

As a Clinic Manager

I want a dashboard showing core operational KPIs

So that I can monitor performance daily.

Acceptance Criteria

- No-show rate displayed by clinic + provider
 - Average wait time visible and filterable
 - Data updates automatically from warehouse
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User Story 3.2 — Build Provider Utilization Heatmap

As a Operations Director

I want provider utilization visualized as a heatmap

So that workload imbalance is immediately visible.

Acceptance Criteria

- Heatmap built in BI tool
 - Ability to filter by clinic, date range
 - Under/over-utilization thresholds clearly highlighted
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User Story 3.3 — Create High-Risk Patient List

As a Front-Desk Coordinator

I want a list of predicted high-risk no-show patients

So that I can proactively make reminder calls or reschedule.

Acceptance Criteria

- List includes name, appointment time, risk score
- Filterable by date, clinic
- Export available (CSV / XLSX)

EPIC 4: Governance, Quality, and Project Enablement

Goal: Ensure clear communication, alignment, and a structured delivery process.

User Story 4.1 — Create Project Governance Structure

As a Project Manager

I want clear roles, responsibilities, and reporting lines

So that the team operates efficiently during the sprint.

Acceptance Criteria

- RASCI created and approved
- Communication plan published
- Daily stand-up schedule set

User Story 4.2 — Define Acceptance Criteria for Final Deliverables

As a Business Analyst

I want well-documented acceptance criteria for each deliverable

So that the team knows what “done” looks like.

Acceptance Criteria

- Criteria documented for model, API, dashboards, schema
- Reviewed by PM + stakeholders
- Added to Trello cards

User Story 4.3 — Prepare Stakeholder Demo & Final Report

As a Project Manager

I want a clear, concise demo + insights report

So that leadership understands the impact of the MVP.

Acceptance Criteria

- Slide deck created
- Insights report drafted
- Demo rehearsal completed